

# CO2xN: Porewater SO<sub>4</sub>/Cl

July 2026 Samples

2026-09-29

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##Add Required Packages

## 0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-09-24" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "July"
Site = "GCRew"
Project_name = "CO2xN"
run_notes = "Had to split 2M July into two runs, computer restarted and halted run.
Needed to re-run standard curve" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/CO2xN_202607_C1_2.txt"
raw_file_name_so4 = "Raw Data/CO2xN_202607_S04_2.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCRew_CO2xN_PW_Processed_C1_S04_202607_2.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCRew", "GCRew_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = here("Porewater QAQC Logs", "GCRew_S04_C1_QAQC_Log.csv")
```

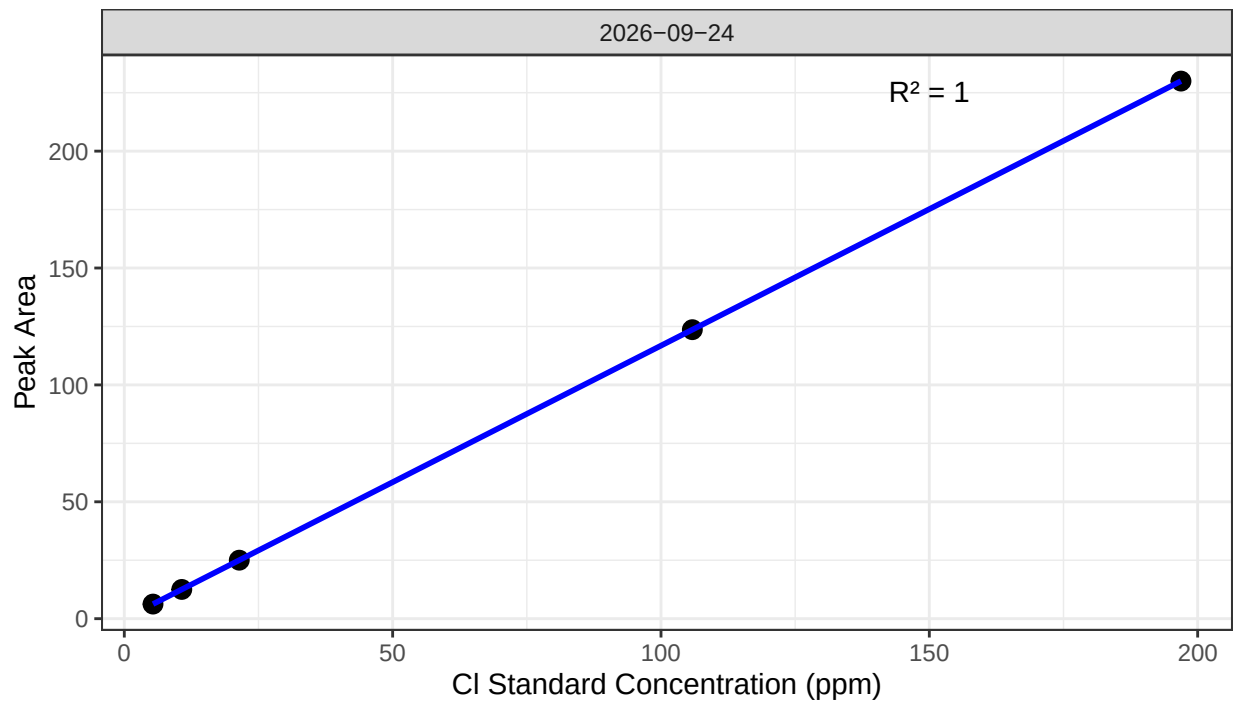
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

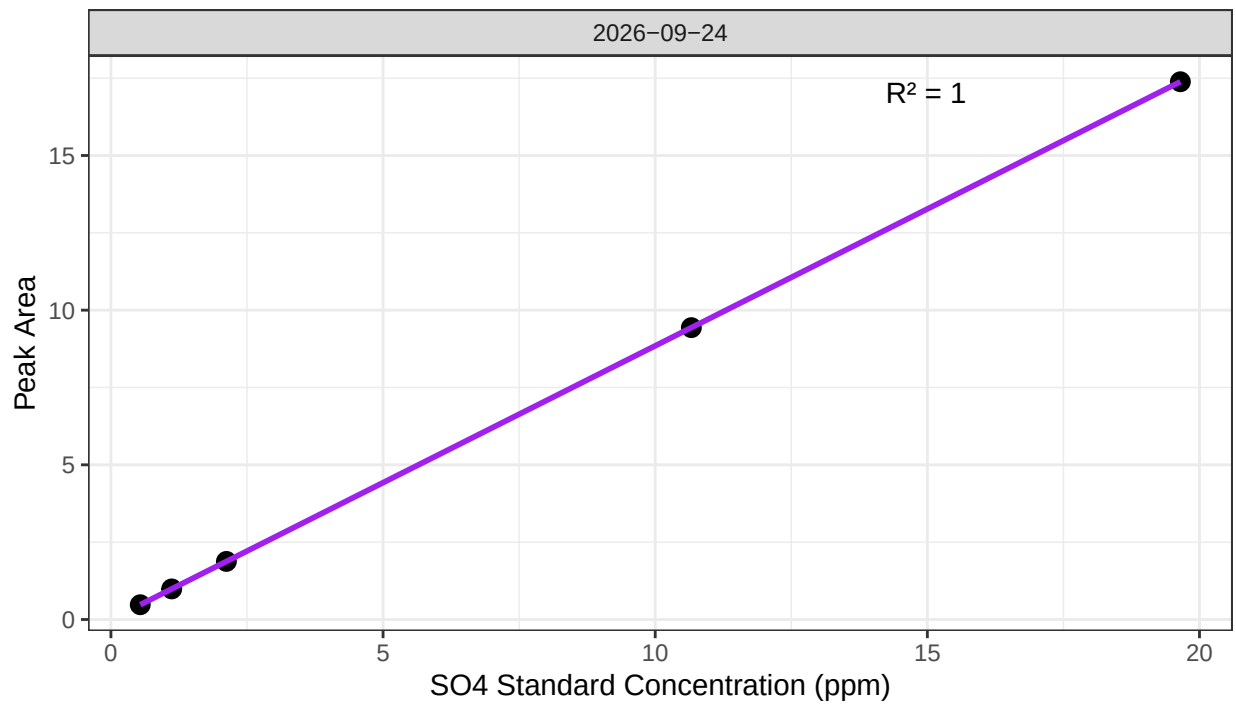
##Import Sample Data

## 0.2 Assess Standard Curves

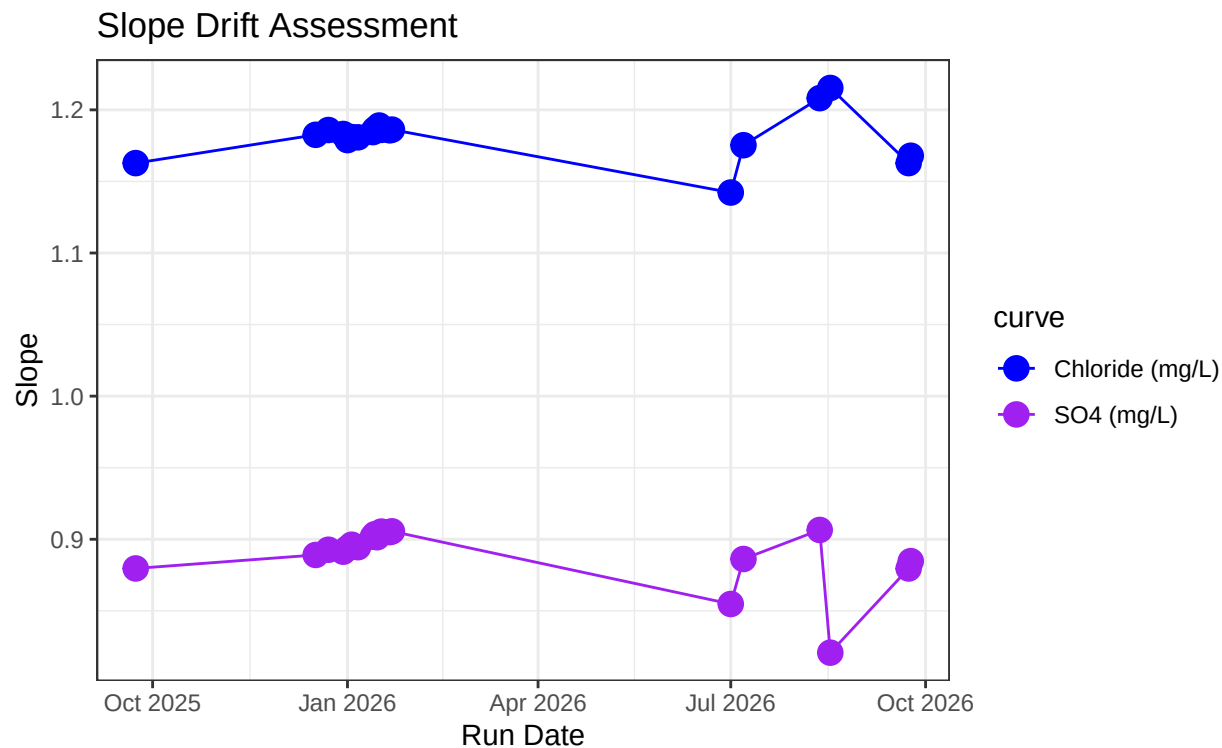
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

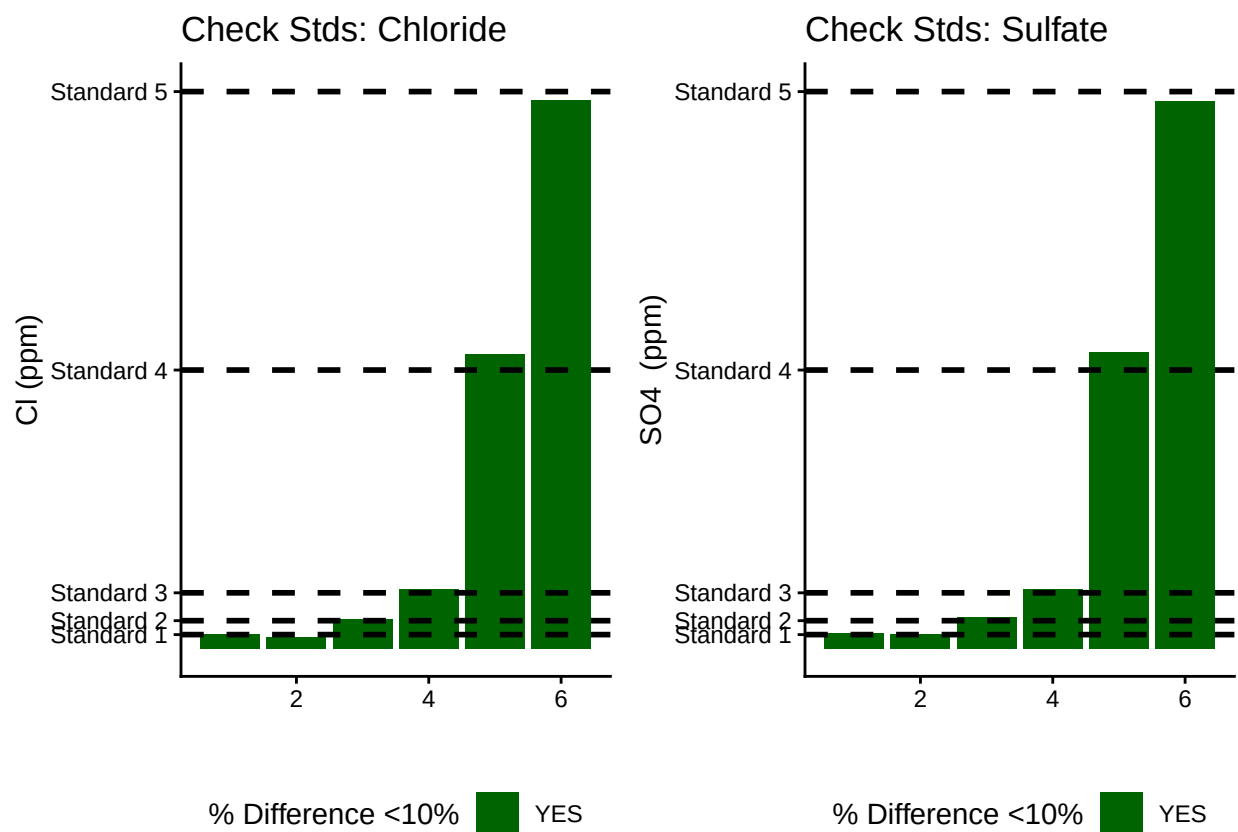
### 0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  4.83  0.736  0.153 Chloride Check Standard RSD within Range - P~
## 2 Standard 2  10.7   NA      NA      <NA>
## 3 Standard 3  21.4   NA      NA      <NA>
## 4 Standard 4  106.   NA      NA      <NA>
## 5 Standard 5  197.   NA      NA      <NA>
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1  0.523  0.0235  0.0449 Sulfate Check Standard RSD within Range --
## 2 Standard 2  1.12   NA      NA      <NA>
## 3 Standard 3  2.12   NA      NA      <NA>
## 4 Standard 4  10.7   NA      NA      <NA>
## 5 Standard 5  19.6   NA      NA      <NA>
```

```
## [1] NA
```

```
## [1] NA
```



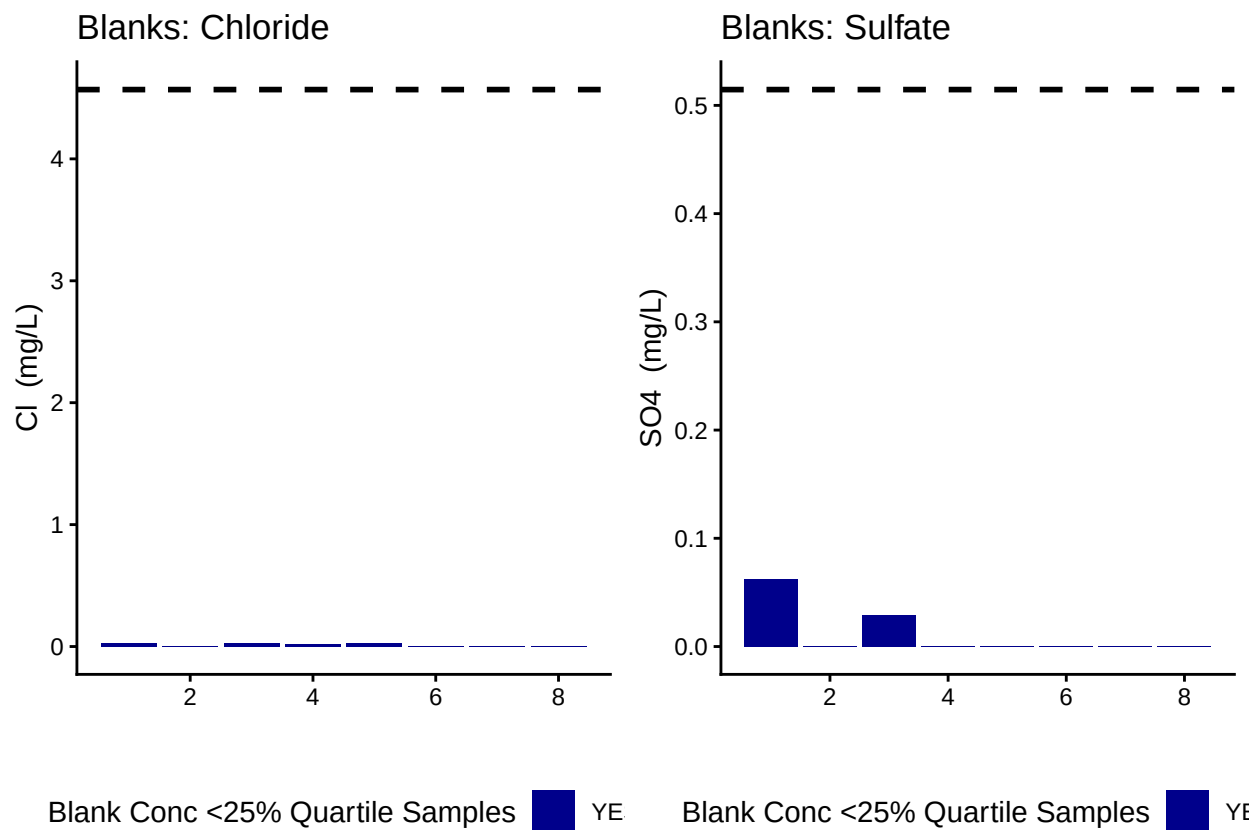
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

#### 0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



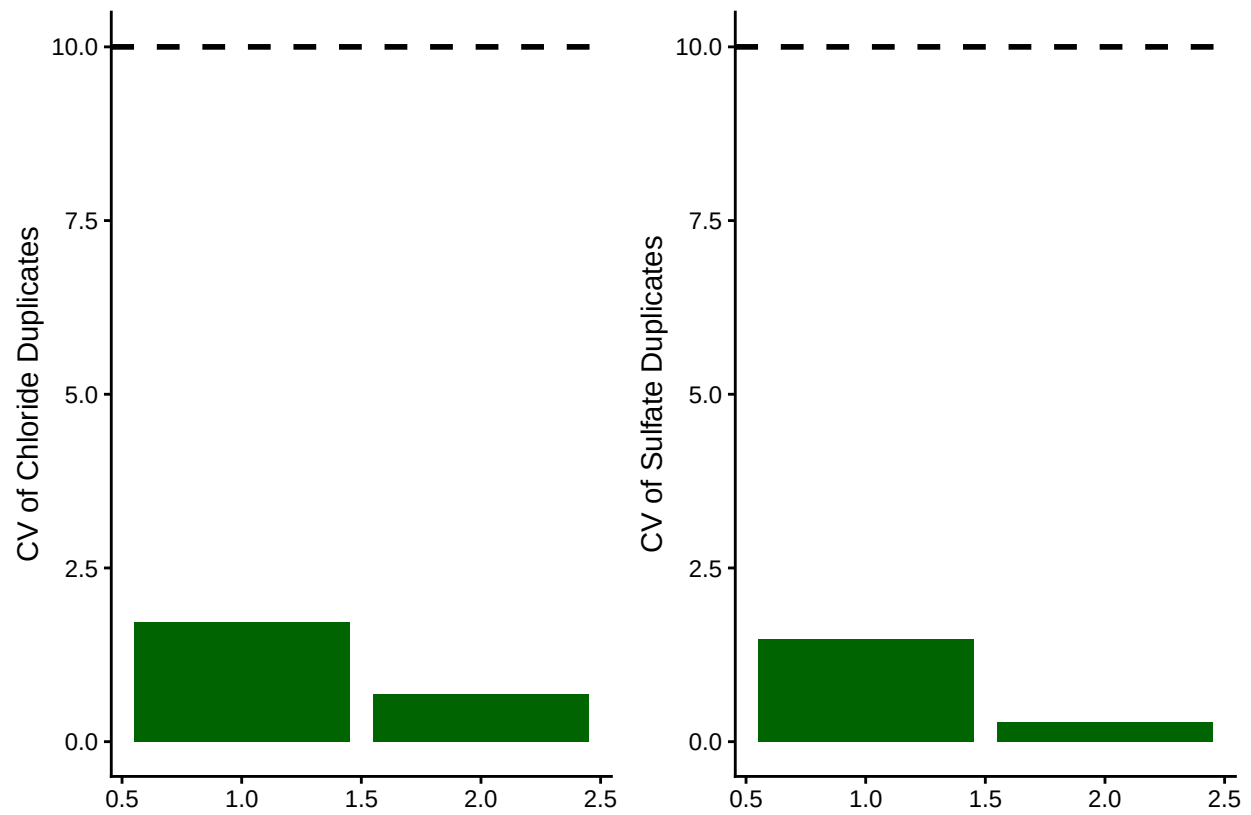
```
## Chloride blanks mean ppm:
```

```
## [1] 0.012875
```

```
## Sulfate blanks mean ppm:
```

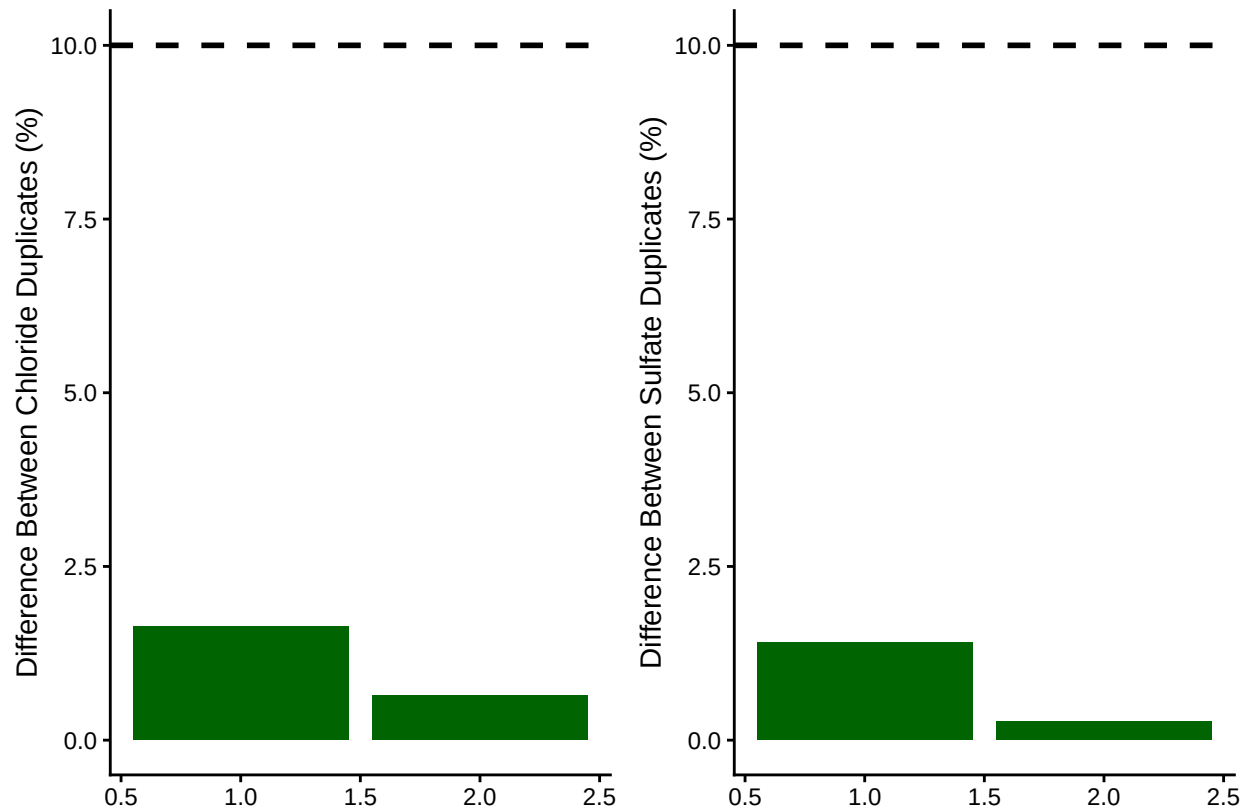
```
## [1] 0.0114875
```

## 0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

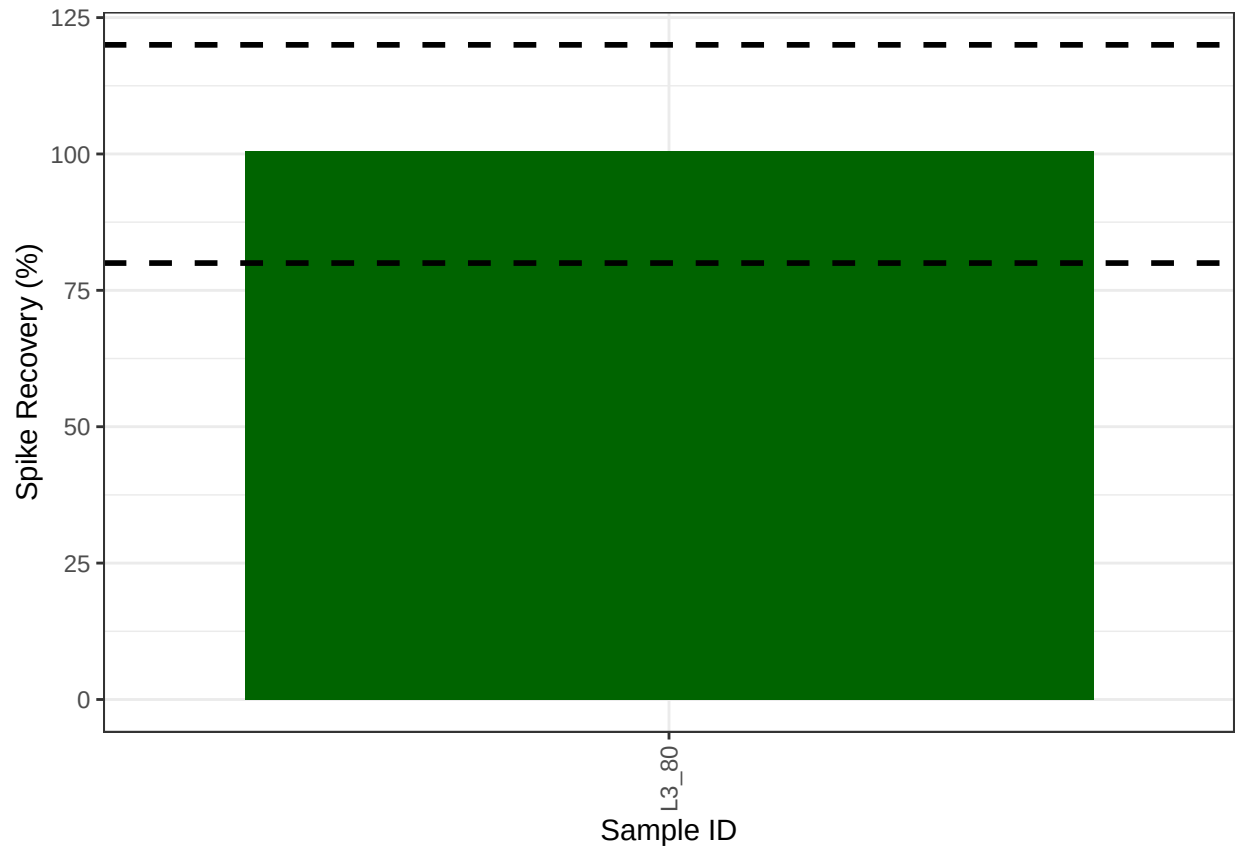
## 0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

## 0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

## 0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

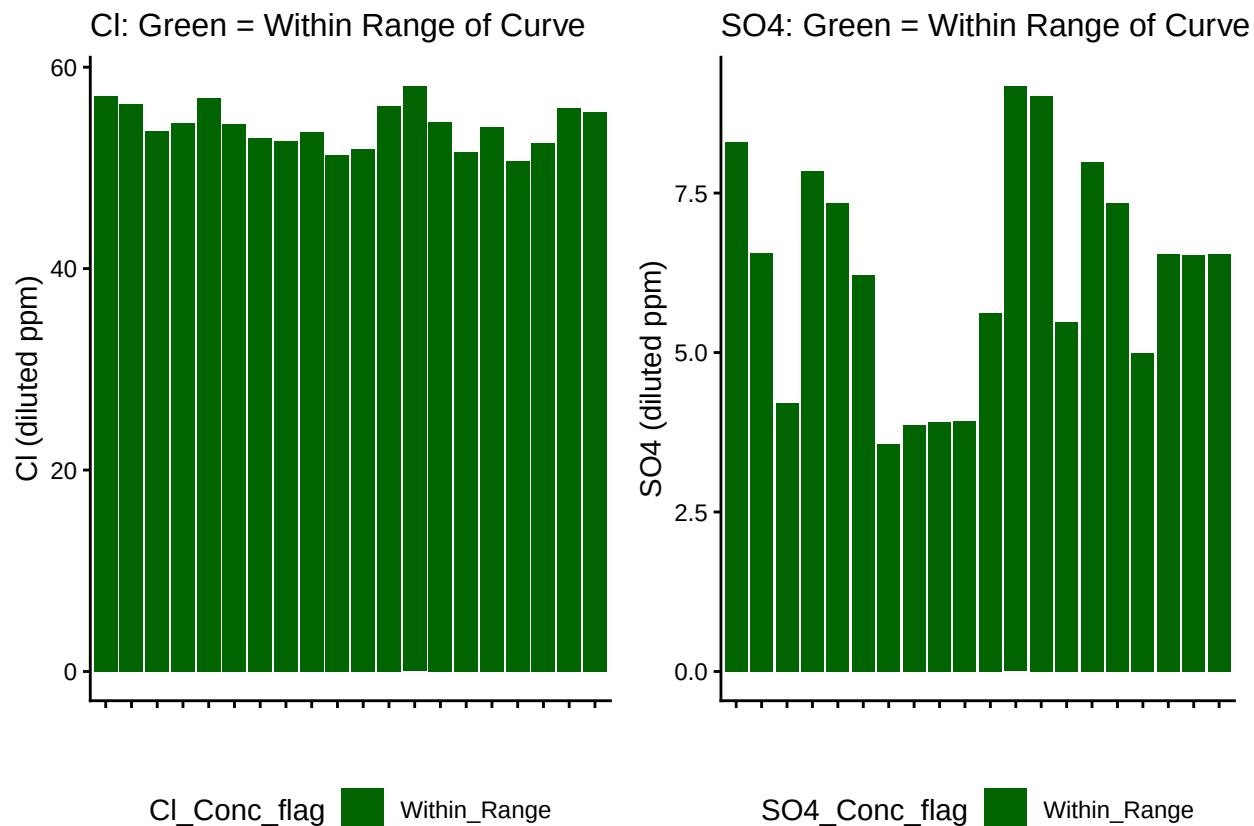


Table 1: SO4 samples

| SO4_Conc_flag | Percent_samples |
|---------------|-----------------|
| Within_Range  | 100             |

Table 2: Cl samples

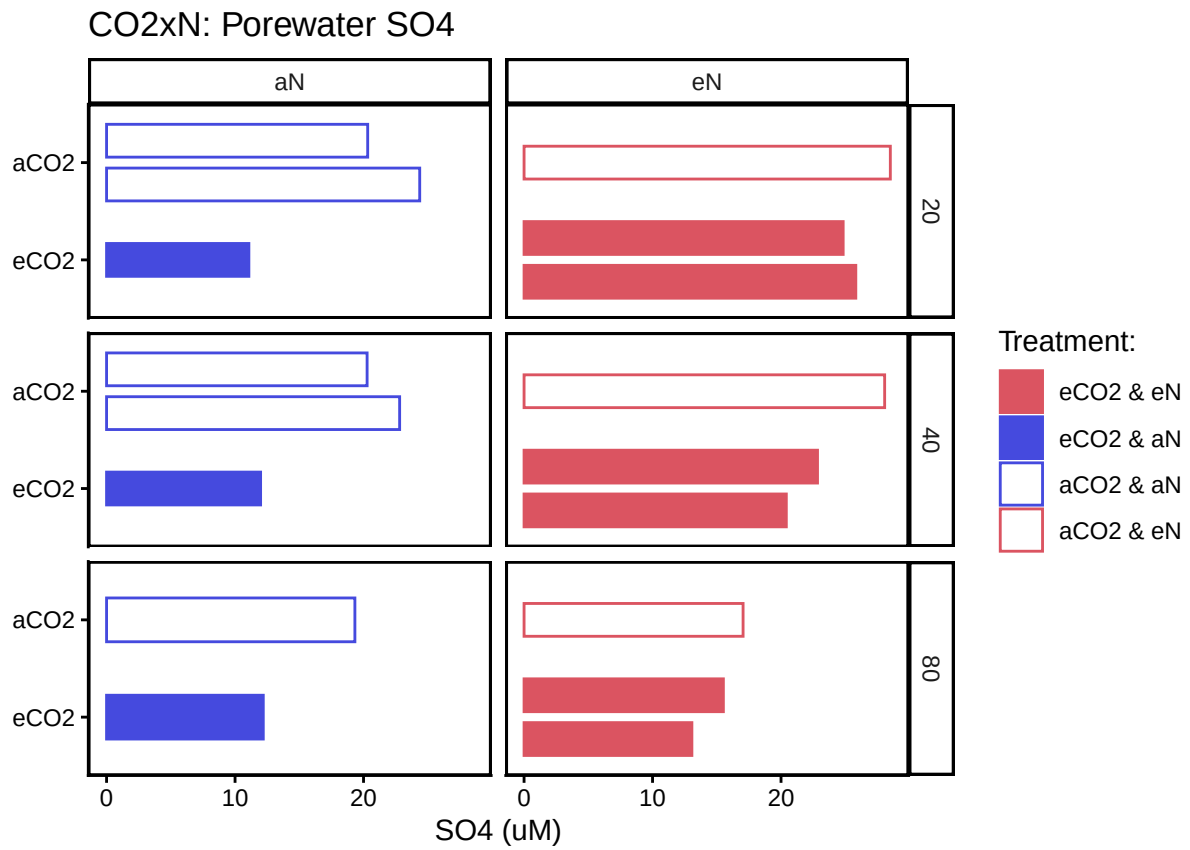
| Cl_Conc_flag | Percent_samples |
|--------------|-----------------|
| Within_Range | 100             |

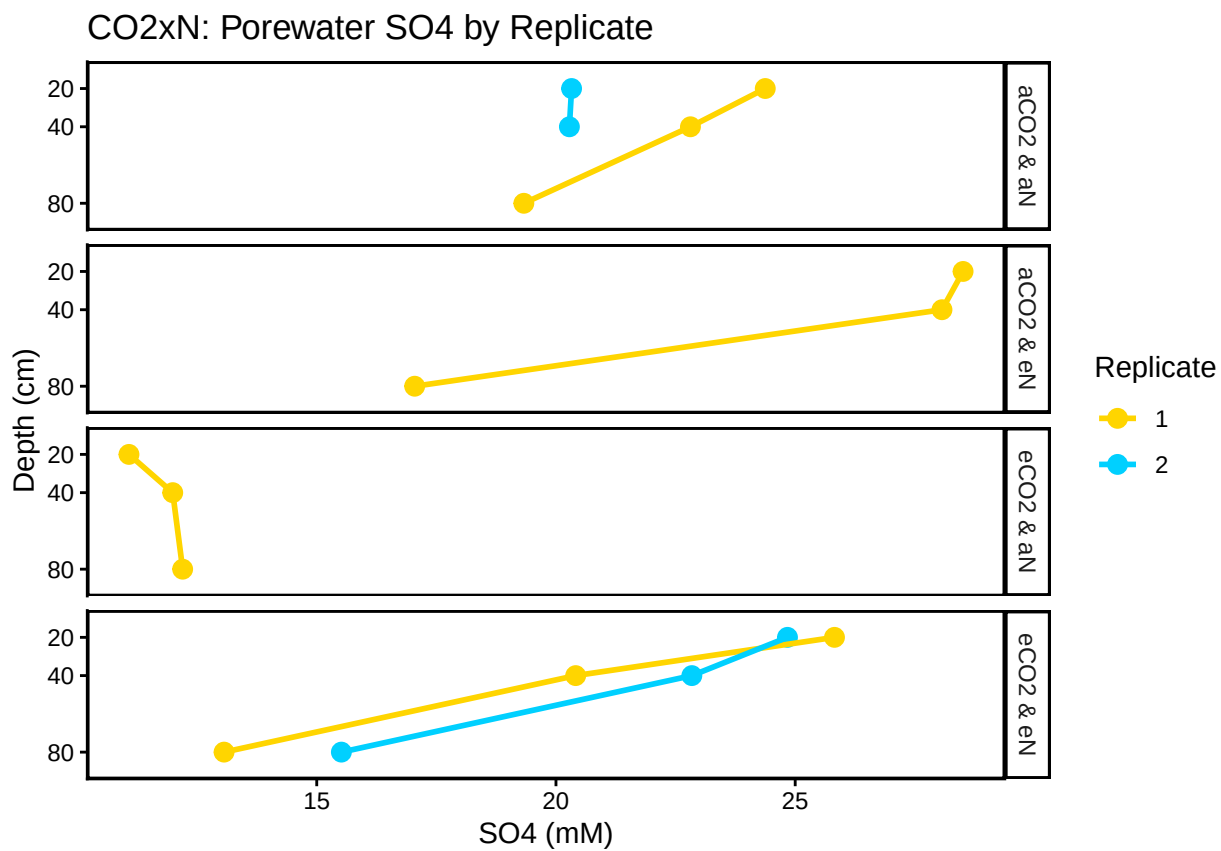
## 0.9 Check to see if samples run match metadata & merge info

## All sample IDs are present in metadata.

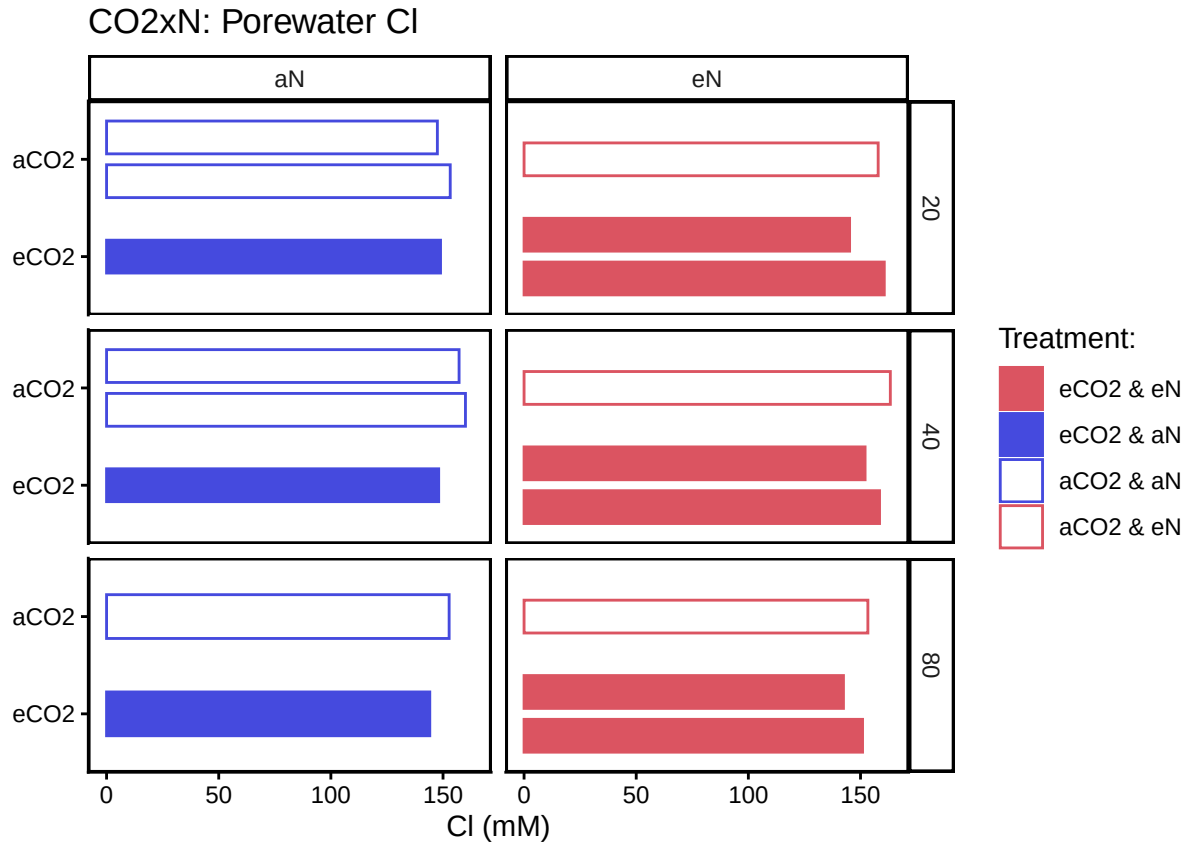
## 0.10 Organize Data

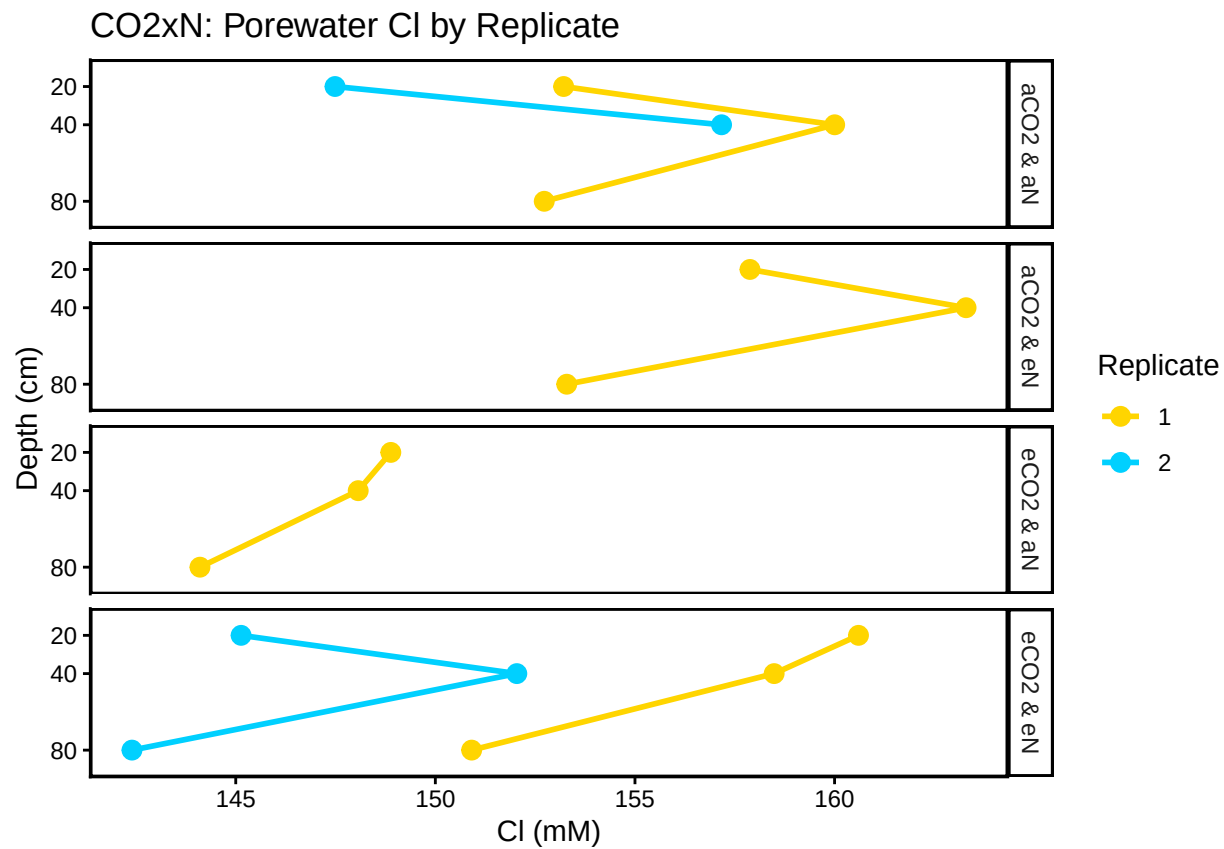
## 0.11 Visualize Data SO4





## 0.12 Visualize Data Cl





### 0.13 Export Processed Data

#end